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PALEOMAGNETISM OF THE 765 MA LUAKELA VOLCANICS IN NORTHWEST ZAMBIA AND IMPLICATIONS FOR NEOPROTEROZOIC POSITIONS OF THE CONGO CRATON

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ABSTRACT. The 765 Ma Luakela volcanics of northwest Zambia are a northeast-trending belt of basaltic to andesitic flows within an essentially undeformed and unmetamorphosed siliciclastic succession that unconformably overlies Neoarchean and Paleoproterozoic rocks of the Congo craton margin. Paleomagnetic analysis of 65 samples from nine sites isolated three magnetization components. Component A, carried mainly by single-domain magnetite, is directed very shallowly to the southeast and northwest, and is interpreted as a primary thermoremanent magnetization. Component B, carried mainly by hematite and oriented mainly shallowly upwards to the southwest, is interpreted to be a secondary overprint, possibly acquired during latest Neoproterozoic (Pan-African) deformation in the Lufilian Arc. A low-stability component C, probably carried by multi-domain magnetite or hematite, or both, is directed very steeply downward, and is similar to Late Paleozoic results from the region. Paleopoles LVA and LVB, corresponding to components A and B, are similar to previous results from mafic rocks of similar age in Tanzania. LVA (40.2°S, 122.0°E) coincides with a reliable pole for the 748 Ma Mbozi syenite-gabbro complex, confirming the position of the Congo craton at 765 to 750 Ma. The secondary LVB pole (38.3°N, 90.9°E), of uncertain age but younger than 765 Ma, coincides with the pole for the *ca.* 795 Ma Gagwe lavas, casting doubt on the assumed primary nature of the Gagwe remanence. Our new data imply that the Congo craton rotated *ca.* 90° clockwise between 750 Ma and the time of B-component acquisition, contrary to previous scenarios in which Congo rotated counter clockwise between *ca.* 800 and 750 Ma.

Key words: Congo craton, Katanga Supergroup, Lufilian Belt, Neoproterozoic, paleomagnetism

INTRODUCTION

The Congo craton (fig. 1) was amalgamated along several orogenic belts during Paleo- and Mesoproterozoic time, and by the Neoproterozoic consisted of the Archean-Paleoproterozoic Angola-Kasai, Bangweulu, Tanzania, Gabon, and São Francisco blocks, and Archean-Paleoproterozoic crust exposed along the northern and northeastern margins of the craton (see summary in De Waele and others, 2008). Along the southern Congo margin, mid-Neoproterozoic rifting and passive margin development were followed by "Pan-African" collisional events during the *ca.* 800 to 530 Ma formation of the supercontinent Gondwanaland (Johnson and others, 2005), within which the Congo craton occupied a central position. However, the positions of the Congo craton during breakup of the Rodinia supercontinent and the subsequent amalgamation of Gondwanaland are very poorly known (Pisarevsky and others, 2003), owing to the small number of reliable, well-dated Neoproterozoic paleomagnetic poles.

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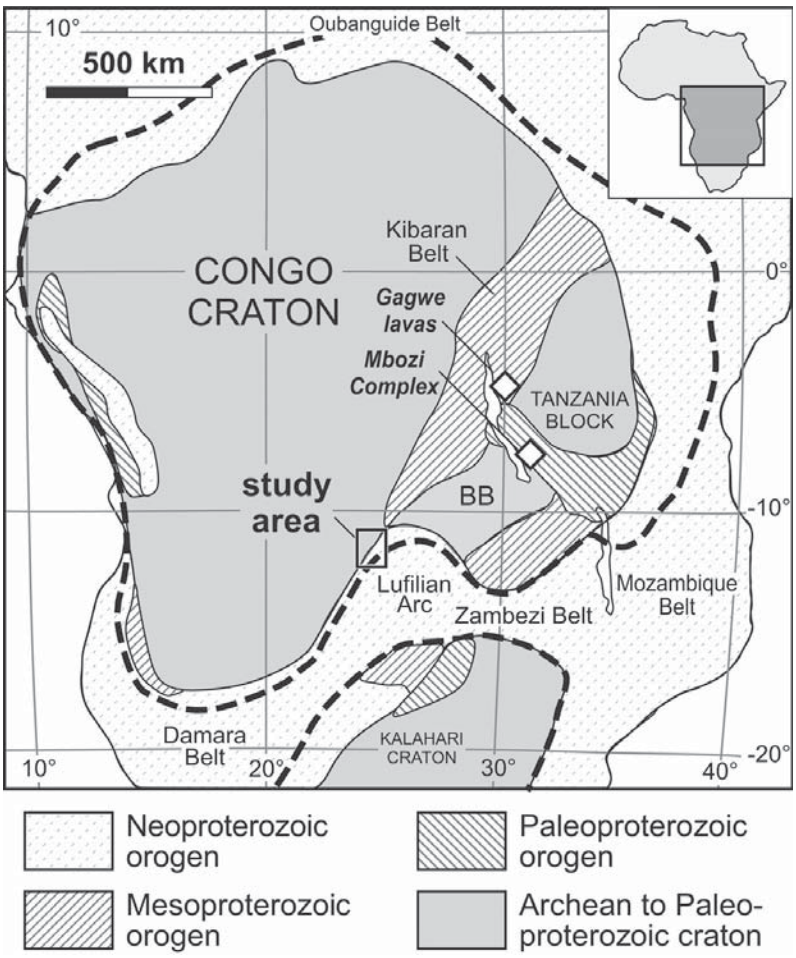


Fig. 1. Simplified geological sketch map of south-central Africa, showing the location of the study area relative to the Congo craton and related tectonic elements. The heavy dashed line shows the approximate limits of the Congo craton in Africa. Also shown are the locations of the Gagwe lavas and Mbozi Complex (Meert and others, 1995). BB, Bangweulu Block.

In this contribution, we present new paleomagnetic results for the Luakela volcanics, a northeast-trending belt of basaltic to andesitic flows in northwest Zambia. The volcanic rocks occur within an autochthonous supracrustal succession that unconformably overlies Archean and Paleoproterozoic magmatic and metamorphic rocks at the southern margin of the Congo craton. These volcanic rocks were identified as an ideal target for paleomagnetic study, having recently been mapped in detail, and dated by SHRIMP U-Pb zircon techniques at *ca.* 765 Ma (Key and Banda, 2000; Key and others, 2001). Comparison of our new data with previous results for the Congo craton requires a reassessment of published models for mid-Neoproterozoic motion of the Congo craton.

GEOLOGICAL BACKGROUND

In the Mwinilunga area of northwest Zambia (fig. 2), the southern margin of the Congo craton is composed of 2560 to 2540 Ma magmatic and metamorphic rocks, intruded by extensive *ca.* 2060 Ma porphyritic granites (Key and others, 2001). These basement rocks are overlain by a northeast-striking volcano-sedimentary succession, correlated with the Katanga Supergroup by Key and others (2001). The succession extends well into the neighboring countries of Angola and the Democratic Republic of

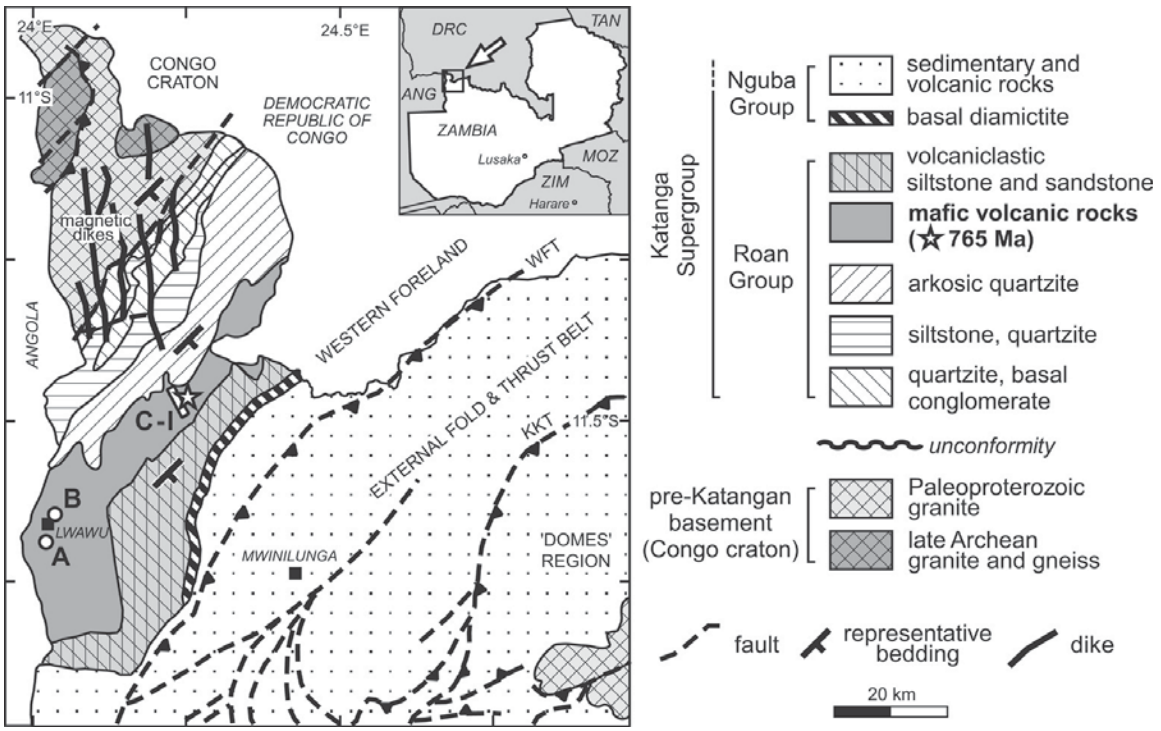


Fig. 2. Geological map of the Mwinilunga area of northwest Zambia (after Key and others, 2001), showing the locations of sites A and B, near Lwawu, and sites C to I, along the Luakela River (fig. 3). The star indicates the approximate location of basaltic andesite samples dated by Key and others (2001). KKT, Kakoma-Kamano Thrust; WFT, Western Foreland Thrust. Stratigraphic column is based on Key and others (2001) and Cailteux and others (2005). Inset shows the location of the study area (arrow) in northwestern Zambia. ANG, Angola; DRC, Democratic Republic of Congo; MOZ, Mozambique; TAN, Tanzania; ZIM, Zimbabwe.

Congo. Fluviatile quartzites and siltstones of the Roan Group are correlated with the lower parts of the Katanga Supergroup and are in unconformable contact with basement rocks. These siliciclastic rocks are overlain by the Luakela volcanics, a sequence of basaltic to andesitic flows correlated with the upper part of the Roan Group, named the Mwashia Subgroup. Zircons from two samples of basaltic andesite from the Luakela volcanics yielded SHRIMP U-Pb zircon ages of 765 ± 5 and 763 ± 6 Ma (Key and others, 2001). The volcanic rocks are up to 800 m thick and, together with underlying and overlying sedimentary rocks, form a homoclinal sequence that dips shallowly (0-10°) to the southeast (Key and others, 2001). Dolerite stocks and north-trending dikes (fig. 2) with strong magnetic signatures intrude the Roan Group and may be feeders to the Luakela volcanics (Key and Banda, 2000). The Luakela volcanics are overlain by about 200 m of magnetic siltstones derived from weathering of the volcanic rocks, followed by an unknown thickness of poorly-exposed diamictite, correlated by Key and Banda (2000) and Key and others (2001) with the "Grand Conglomerat," a widespread glaciogenic unit, considered to be of Sturtian age, that defines the base of the Nguba Group.

To the southeast of the study area, Katangan rocks are strongly deformed and metamorphosed within the Lufilian Belt (fig. 1), a *ca.* 550 to 530 Ma (Pan-African) arcuate thrust and fold belt related to collision of the Congo and Kalahari cratons (Porada and Berhorst, 2000; Key and others, 2001; Wendorff, 2003). Essentially undeformed Katangan strata in the Mwinilunga area rest on the Congo craton margin and are part of the western foreland to the Lufilian Belt. Katangan sedimentary rocks in the Mwinilunga area are essentially unmetamorphosed, although the Luakela

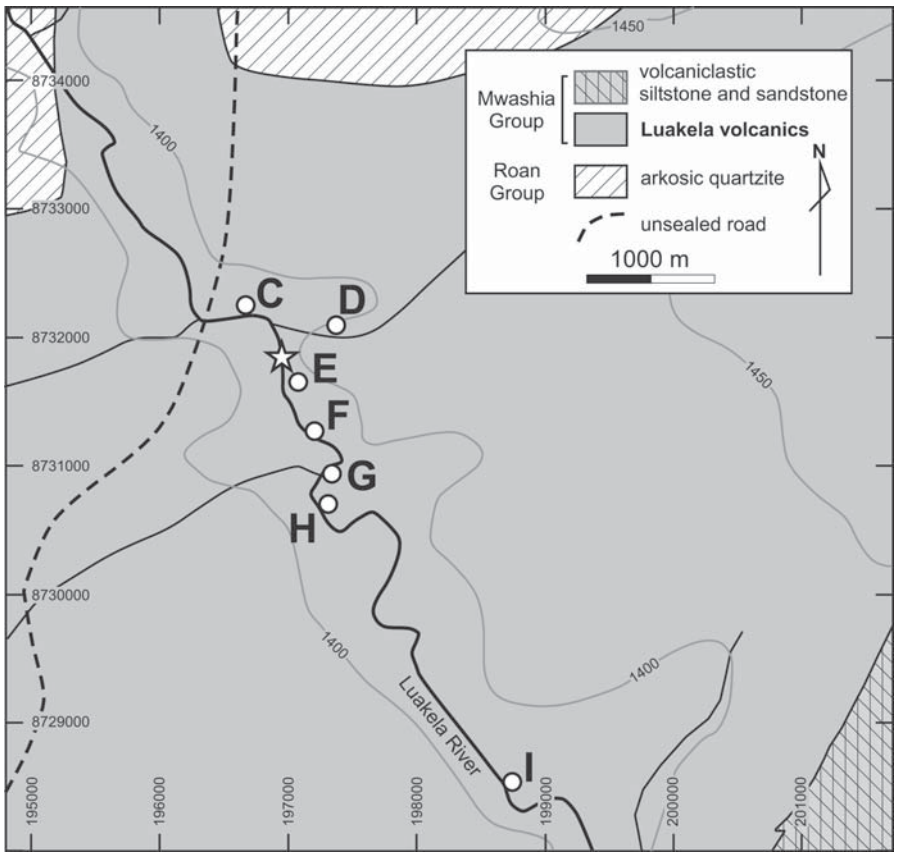


Fig. 3. Sample sites C to I along the Luakela River. The star indicates the approximate location (R. M. Key, pers. comm., 2009) of rocks dated at *ca.* 765 Ma by Key and others (2001). Numbered gray curves are elevation contours; coordinates refer to UTM zone 35L.

volcanics are described as being strongly altered, with primary minerals partially replaced by low-temperature, hydrated phases (Key and Banda, 2000; Key and others, 2001).

PALEOMAGNETIC SAMPLES AND ANALYTICAL TECHNIQUES

Samples of the Luakela volcanics were collected from nine sites in two areas. Two sites (A & B, fig. 2) were located near Lwawu, close to the Angola border. Site A (11.6845°S, 24.0371°E) was located in large outcrops on the southwest side of Sakapochi School, and site B (11.6244°S, 24.0595°E) encompassed about 70 m of outcrop along the east bank of the Lwawu River immediately west of the Lwawu Mission. Seven sites (C to I, fig. 3) were located in a section exposed intermittently for about 8 km along the Luakela River (between 11.4559°S, 24.2201°E and 11.4906°S, 24.2396°E). The river cuts across the strike of the volcanic belt (fig. 3), and the section includes the exposures of basaltic andesite dated at *ca.* 765 Ma by Key and others (2001). Between five and 12 cores, 25 mm in diameter and 5 to 10 cm long, were collected at each site using a petrol-powered drill, and oriented with magnetic and solar compasses. Sun readings were used for orientation of all cores, owing to the highly magnetic nature of many outcrops. Geographic coordinates were determined using a GPS receiver. Other than possible flow-top features (amygdale horizons, breccias), no other primary volcanic features were recognized in the field, and the subhorizontal attitude and inadequate exposure of the rocks prevented us from determining structural attitudes or whether more than one flow was sampled at each site.

The sampled volcanic rocks are fine- to medium-grained, typically massive, and locally banded, amygdaloidal, or feldspar-phyric. Alteration is ubiquitous in the volcanic rocks, with sericite after plagioclase, chlorite replacing clinopyroxene, and secondary quartz, calcite, and epidote (Key and Banda, 2000). Many outcrops are pervasively hematized, and the rocks are purple-red in color. In some cases, hematite occurs in irregular bands, suggestive of secondary alteration processes. Two main magnetic minerals, magnetite and hematite, are distinctive in thin sections. Magnetite is typically bimodal in size, occurring as spectacular cruciform skeletal octahedra up to 0.2 mm across (figs. 4A and 4B), and as locally abundant grains typically <0.01 mm across but up to 0.05 mm. The large cruciform crystals are clearly primary, having crystallized before silicate grains, and consist of titanium-poor magnetite transected by sparse to abundant ilmenite oxidation-exsolution lamellae, the latter variably altered to leucoxene (figs. 4A and 4B). Smaller, secondary magnetite grains are irregular to euhedral, mainly interstitial to altered silicate grains, and do not exhibit exsolution lamellae. Hematite is locally pervasive in the groundmass and also forms thin rims around some magnetite grains. In some samples, a heavy dusting of fine-grained hematite is concentrated around vesicles subsequently filled with chlorite (fig. 4C).

In the laboratory, each core sample was cut into 22-mm-long specimens, which were demagnetized thermally using *Magnetic Measurements* MMTD1 furnaces, and by alternating magnetic fields (AF) via a 2G600 automated degaussing system within the *2G-Enterprises* 2G755R cryogenic magnetometer used for all remanence measurements. One specimen from each sample was subjected to stepwise AF demagnetization up to 120 mT, or until the magnetization became unstable. A second specimen from each sample was demagnetized thermally up to 700 °C. Numerous hematite-bearing specimens were demagnetized initially by AF to 120 mT, and then thermally up to about 700 °C. Detailed susceptibility versus temperature measurements were obtained for selected samples with a MS2 susceptibility meter. Magnetization vectors were isolated using Principal Component Analysis (Kirschvink, 1980). Results for specimens from the same core sample were averaged and sample means given unit weight in statistical calculations; paleopoles were calculated from site mean directions.

PALEOMAGNETISM

The intensity of natural remanent magnetization (NRM) of the studied rock samples ranges from 0.1 to 100 A/m, and the magnetic susceptibility (k) from 0.1 to 100 ($\times 10^{-5}$) SI units. The Koenigsberger ratio (Q-ratio), of remanent to induced magnetization, is high (1.4-129), suggesting high paleomagnetic stability. However, some samples with very high Q values also have very high NRM intensities (>60 A/m) and yield remanence directions that are typically steep and discordant to those of other samples from the same site. These characteristics are indicative of lightning-induced isothermal remanence (IRM, Graham, 1961), and we excluded results from several such samples from subsequent calculations. The variation of low-field susceptibility with temperature (fig. 5) shows well-defined Hopkinson peaks close to 580 °C, indicating the presence of fine-grained (single-domain, SD) magnetite (Schmidt, 1993). The concentration of magnetite in these rocks is sufficiently high that magnetic susceptibility does not exceed the level of noise after demagnetization at temperatures above 580 °C. However, unblocking temperatures up to 700 °C, as well as field and microscopic observations, clearly indicate the presence of hematite and its important role in the paleomagnetic signature of these rocks.

Thermal and AF demagnetization of 65 samples (109 specimens) from nine sites isolated three magnetization components: A, B, and C (fig. 6). Some samples contain only component A, others only component B, and some contain both A and B. Where both A and B are present, the magnetization vector rotates from an A direction towards B during progressive demagnetization. Sites A, B, D, E, and I possess both components

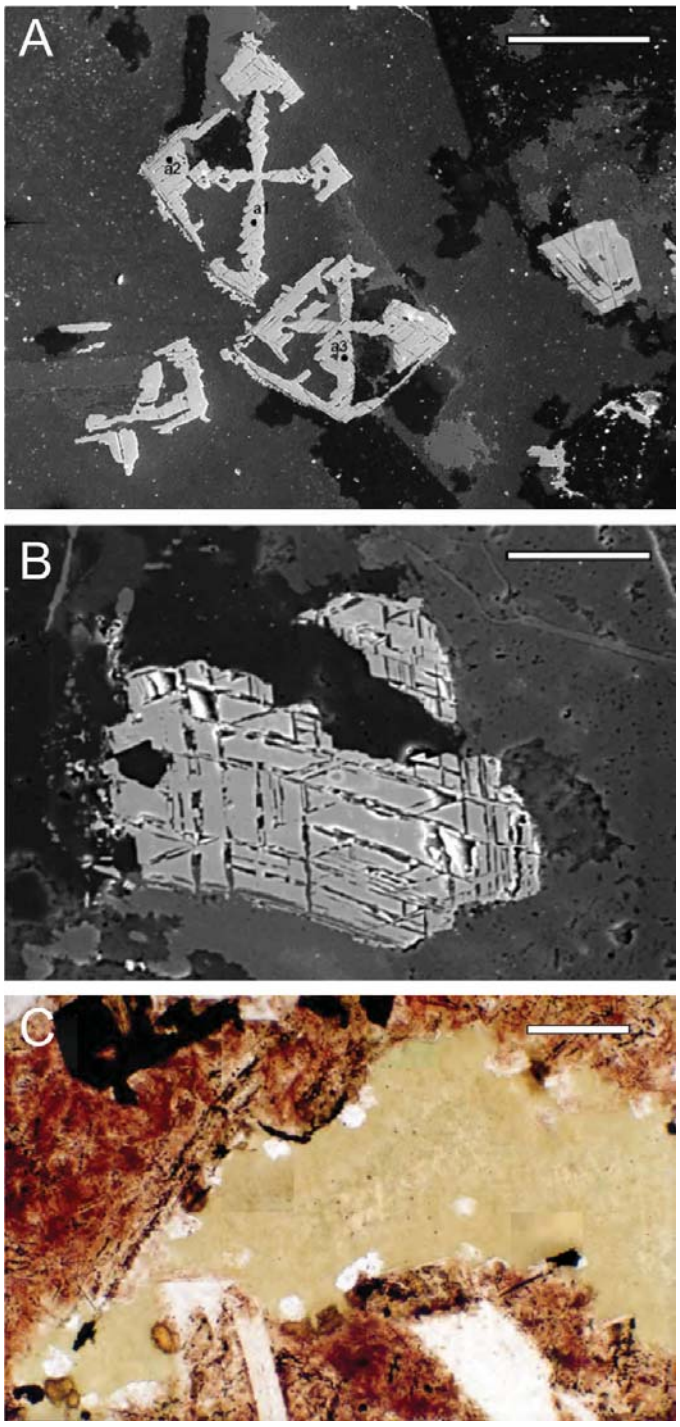


Fig. 4. Photomicrographs showing petrographic features of paleomagnetic samples from the Luakela volcanics. (A) Reflected-light image of cruciform magnetite crystals at site A. (B) Reflected-light image of altered ilmenite lamellae within a low-Ti magnetite crystal, site B. (C) Transmitted-light image of chlorite-filled amygdale, around which the rock is strongly hematized, site E. The scale bar in each image is 200 microns.

A and B, whereas sites C and H yield only component A, and sites F and G only component B. Component C is found in samples from sites C, F, H, and I.

Component C has subvertical inclination and is generally of low stability (figs. 6B and 7; table 1). The C-component is typically destroyed during the first few steps of AF

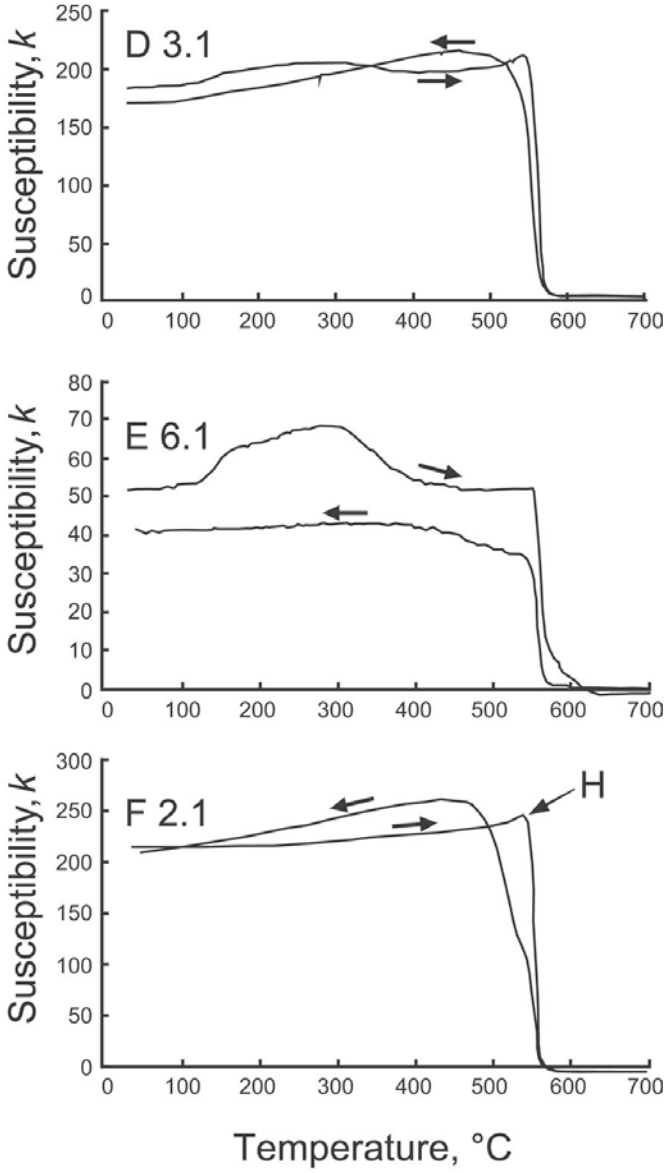


Fig. 5. Variation of low-field susceptibility with temperature for specimens from sites D, E, and F. The pronounced shoulders on the heating curves at about 580 °C, known as Hopkinson peaks (H), indicate the presence of highly stable fine-grained (single-domain) magnetite.

demagnetization, and is probably carried by multidomain (MD) magnetite, a suggestion supported by the petrographic observation of large, variably-exsolved magnetite crystals (figs. 4A and 4B). Steep magnetizations in several specimens were stable but associated with very high NRM intensities (>50-60 A/m), and are attributable to lightning effects.

Component A is present in seven sites, is of dual polarity, and is directed very shallowly to the northwest and southeast (fig. 7, table 1). It is typically isolated at temperatures between 350 and 580 °C (fig. 6A), or during AF treatment at 80 to 120 mT. Owing to overlapping blocking temperature spectra, the A-component in many specimens is contaminated by a strong C-component contribution. Component A directions pass the reversal test of McFadden and McElhinny (1990) at the sample level with R_c classification (the angle between mean normal and reverse directions is 166.4°). At least seven flows are represented at the seven sites, and the data encompass

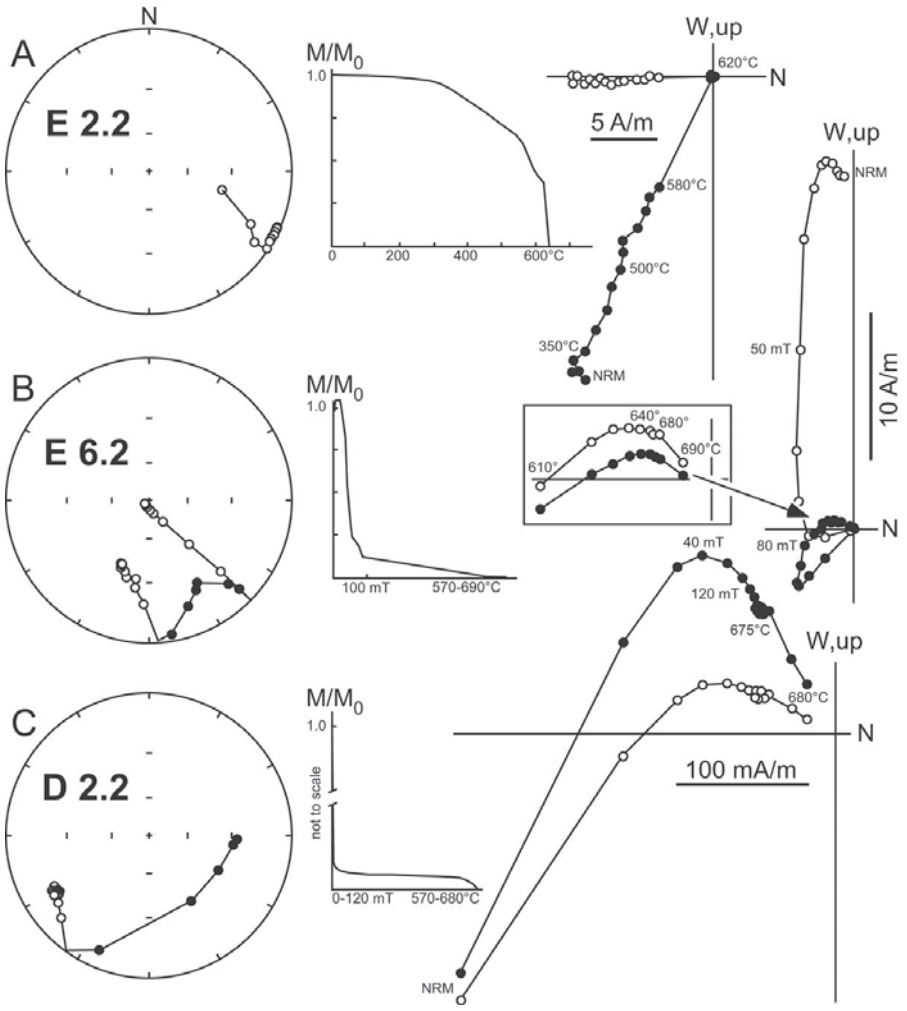


Fig. 6. Representative examples of demagnetisation behaviour. In orthogonal plots, open (closed) symbols show magnetisation vector endpoints in the vertical (horizontal) plane; curves show changes in intensity during demagnetisation (M/M_0, intensity/initial intensity). Equal-area stereographic projections show upwards- (downwards-) pointing paleomagnetic directions with open (closed) symbols.

at least one magnetic field reversal, hence the results are likely to have averaged secular variation. The tilt-corrected A-component paleopole (LVA) is located at 40.2°S, 122.0°E (table 1).

Component B is directed mainly shallowly upward to the southwest (figs. 6C and 7, table 1), essentially orthogonal to the component A direction. AF demagnetization up to 120 mT was generally insufficient to isolate the B-component, and subsequent thermal demagnetization to about 700 °C was required (figs. 6B and 6C). Application of the reversal test to B-component directions is indeterminate (classification R_0). It appears that at least one magnetic field reversal occurred during their acquisition; hence the results are likely to have averaged secular variation. The tilt-corrected B-component paleopole (LVB) lies at 38.3°N, 90.9°E (table 1).

DISCUSSION

Component A is interpreted as a primary thermoremanent magnetization, because it is carried in part by single-domain magnetite (for example, in smaller Ti-poor exsolution lamellae in primary magnetite), does not resemble younger magnetizations from the Congo craton, and because the rocks have not been thermally metamor-

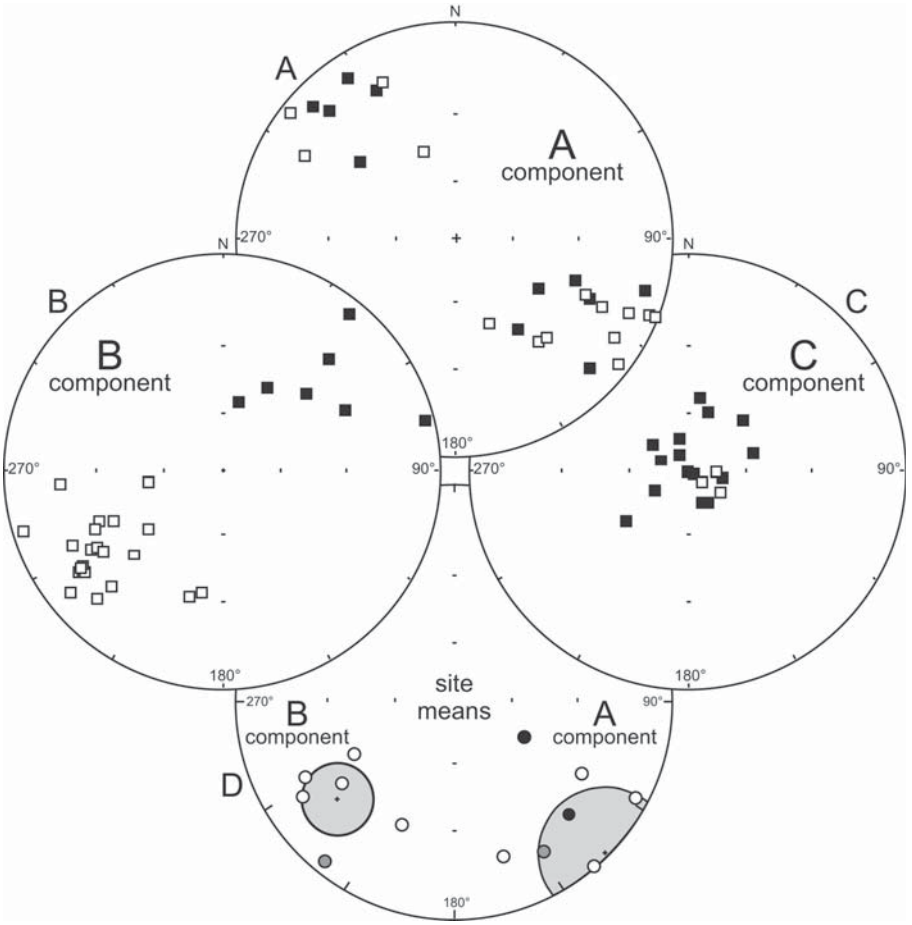


Fig. 7. (A-C) Sample directions defining remanence components A, B, and C, and (D) the means of site directions for A and B components. Refer to fig. 6 for explanation. Gray symbols in (D) indicate mean directions inverted through the origin.

phosed. Single-domain magnetite is highly stable magnetically, requiring heating close to its Curie temperature of about 580 °C to reset its remanence (Pullaiah and others, 1975; Walton, 1980). Component B is carried mainly by hematite, and also based in part on petrographic observations, is interpreted as a younger overprint of unknown

Table 1

Mean paleomagnetic directions and paleopoles for the Luakela volcanics

Component	B/N/n	tilt correction	*D* (º)	*I* (º)	*k*	α_{95} (º)	Pole	Plat (ºN)	Plong (ºE)	*Dp* (º)	*Dm* (º)
A	7/25/40	corrected	130.8	3.3	10.2	19.9	LVA	-40.2	122.0	10.0	19.9
		in situ	130.8	-1.7				-39.6	125.1		
B	7/29/32	corrected	227.1	-20.0	22.2	13.1	LVB	38.3	090.9	7.2	13.7
		in situ	228.9	-19.7	36.8				092.0		
C	4/16/22	in situ	002.9	81.1	18.5	21.9	LVC	5.7	025.0	40.9	42.3

B/N/n, number of sites/samples/specimens; tilt correction refers to bedding attitude of 5° towards 135°; *D, I*, declination, inclination; *k*, Fisher (1953) precision parameter; α_{95}, semi-angle of the 95% cone of confidence about the mean direction; Plat, Plong = latitude, longitude of the paleopole; *Dp*, *Dm*, semi-axes of the oval of 95% confidence about the pole.

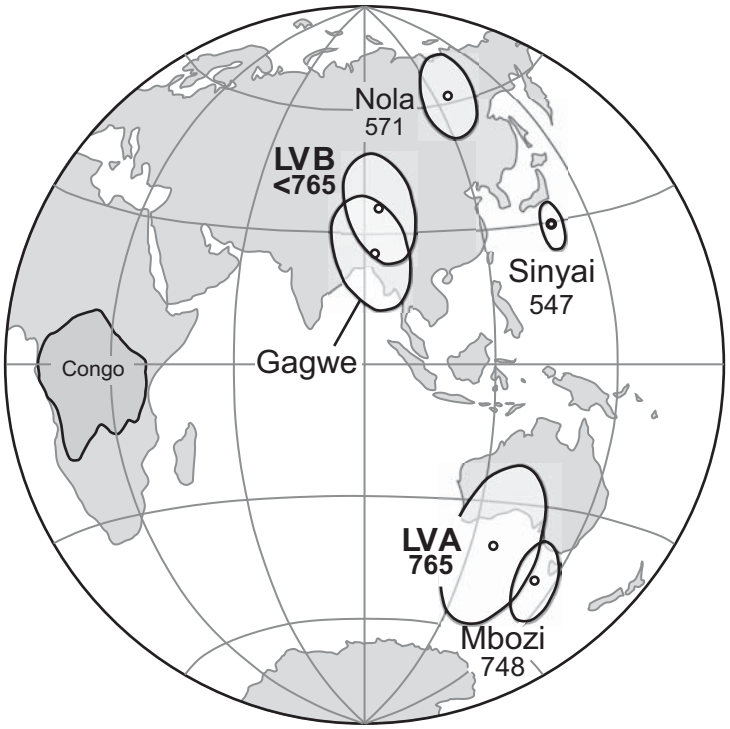


Fig. 8. Paleopoles for the Congo craton from this study and previous investigations. Ages are in million of years (Ma). Gagwe lavas and Mbozi Complex poles are from Meert and others (1995), the Sinyai metadolerite pole is from Meert and Van der Voo (1996), and the Nola dikes pole is from Moloto-A-Kenguemba and others (2008).

age. The C-component paleopole (LVC, table 1) is similar to Early Carboniferous Gondwana poles (McElhinny and others, 2003), and we attribute the C-component to Late Paleozoic remagnetisation and/or lightning strikes.

Paleopoles LVA and LVB, corresponding to components A and B, are more than 80° apart, and similar to previous results from southern Tanzania (fig. 8). Both poles are corrected for the tilt of bedding (table 1). Although it is not clear whether or not LVB should be corrected, the bedding correction is sufficiently small (5°), that it makes no significant difference.

LVA coincides with a reliable paleopole for the Mbozi gabbro-syenite complex (Meert and others, 1995), for which several K-Ar results are within uncertainty of an ion microprobe (SHRIMP) U-Pb zircon age of 748 ± 6 Ma (Mbede and others, 2004). The LVA and Mbozi poles place the Congo craton at the equator at 765 to 750 Ma (fig. 9A). Confirmation that the Congo craton occupied equatorial latitudes at this time implies that the "Grand Conglomerat" diamictite, and other Sturtian glaciogenic rocks in the Congo craton, represent a low-latitude glaciation.

LVB falls within uncertainty of a paleopole for the Gagwe lavas (Meert and others, 1995), for which there is an Ar-Ar plagioclase age of 795 ± 7 Ma, and several K-Ar ages between 815 and 800 Ma that are thought to reflect excess Ar (Deblond and others, 2001). However, our results indicate that the LVB pole is younger than the 765 Ma LVA pole, because the B-component remanence is carried by secondary hematite. Therefore, unless the Tanzania block has moved significantly relative to the Angola-Kasai block since at least 800 Ma, either the 795 Ma Ar-Ar age does not accurately reflect the time of magnetization, or the Gagwe lavas pole represents a younger overprint. There is evidence in the Ar-Ar data for alteration-related isotopic resetting of these rocks at c. 570 to 540 Ma (Deblond and others, 2001).

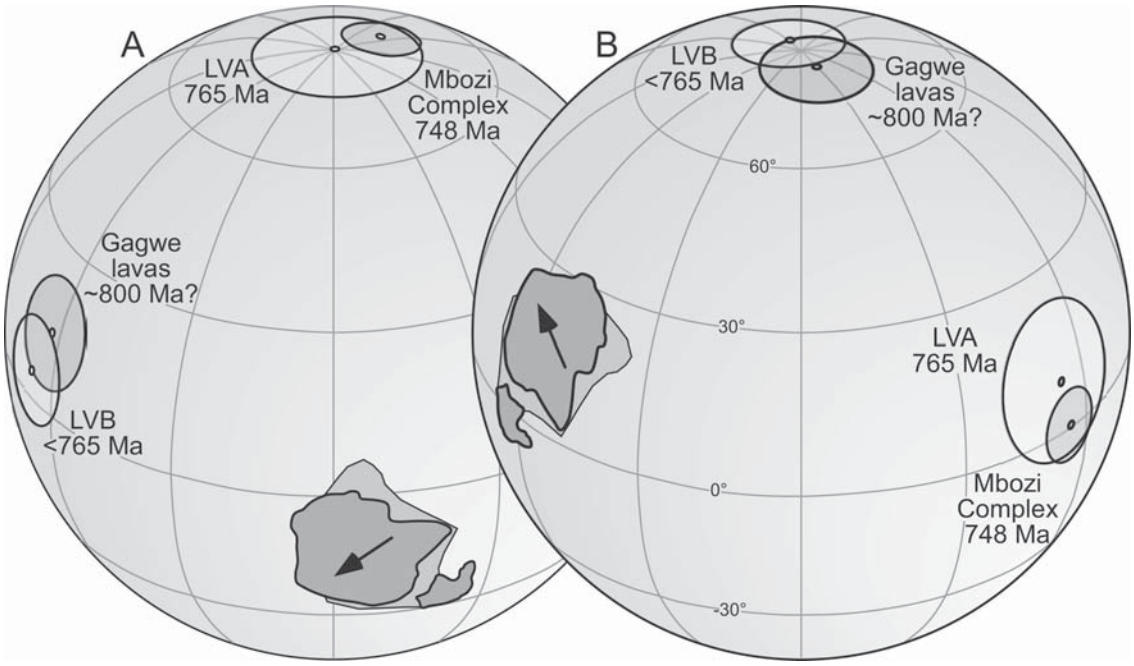


Fig. 9. Positions of the Congo craton: (A) at 765 Ma, and (B) at the time of acquisition of the B-component magnetization.

At the time the B-component remanence of the Luakela volcanics was acquired, the LVB and Gagwe poles place Congo near the equator, oriented about 110° clockwise compared to its position at 765 to 750 Ma (fig. 9B). The LVB and Gagwe poles are about 30° away from an overprint pole, the age of which is estimated to be 571 ± 6 Ma based on Ar-Ar analysis of amphibole from the Nola dike swarm of the northern Congo margin (Moloto-A-Kenguemba and others, 2008), and about 50° away from an overprint pole, dated by Ar-Ar on biotite at 547 ± 4 Ma, for the Sinyai metadolerite dike within the Mozambique Belt of Kenya (Meert and Van der Voo, 1996). However, both of these poles represent secondary overprints from metamorphosed rocks, and the Ar-Ar dates may not accurately reflect the time of remanence acquisition. Together with the evidence for isotopic resetting within the Gagwe lavas, the rough proximity of these poles nevertheless suggests that the LVB and Gagwe magnetizations might have been acquired in the latest Neoproterozoic, during the *ca.* 570 to 530 Ma culmination of Pan-African tectonothermal events (De Waele and others, 2008).

Identification of the Gagwe lavas paleopole as a post-750 Ma overprint means that previous models (for example, Meert and others, 1995; Pisarevsky and others, 2003) which invoked a 90° counter clockwise rotation of Congo in the 50 Ma interval between 800 and 750 Ma are no longer supported. Instead, the Congo craton rotated in a clockwise sense between 750 Ma and the time of B-component acquisition.

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